Chapter 11 • Loci and Construction

# Solved Review Exercise 11

## MCQs

**1. A triangle can be constructed if the sum of any two sides is ___ the third side.**
a) less than
b) greater than
c) equal to
d) greater than or equal to
**Ans: b**
The strict triangle inequality requires greater than.

**2. An equilateral triangle:**
a) can be isosceles
b) can be right-angled
c) can be obtuse-angled
d) has each angle 50°
**Ans: a**
It is a special isosceles triangle with three equal sides.

**3. If two angles sum to less than 90°, the triangle is:**
a) equilateral
b) acute-angled
c) obtuse-angled
d) right-angled
**Ans: c**
The third angle is greater than 90°.

**4. The segment joining a side midpoint to the opposite vertex is a:**
a) median
b) perpendicular bisector
c) angle bisector
d) circle
**Ans: a**
This is the definition of a median.

**5. The angle bisectors of a triangle intersect at:**
a) one point
b) two points
c) three points
d) four points
**Ans: a**
They are concurrent at the incentre.

**6. The locus equidistant from a fixed point is a:**
a) circle
b) perpendicular bisector
c) angle bisector
d) parallel lines
**Ans: a**
All points on the circle have the same radius.

**7. The locus equidistant from two fixed points is a:**
a) circle
b) perpendicular bisector
c) angle bisector
d) parallel lines
**Ans: b**
It is the perpendicular bisector of their joining segment.

**8. The locus equidistant from a fixed line is/are:**
a) circle
b) perpendicular bisector
c) angle bisector
d) parallel lines
**Ans: d**
Two parallels at the fixed perpendicular distance.

**9. The locus equidistant from two intersecting lines is:**
a) circle
b) perpendicular bisector
c) angle bisector

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d) parallel lines
**Ans: c**
The internal and external angle bisectors.

**10. Points farther than 2 km from B lie outside a circle of radius:**
a) 1 km
b) 1.9 km
c) 2 km
d) 2.1 km
**Ans: c**
The bounding circle is centred at B with radius 2 km.

**2. Construct a right triangle with sides 6 cm, 8 cm and 10 cm.**
Draw a 6 cm base and construct a perpendicular segment of 8 cm at one endpoint. Join the free endpoints.
The hypotenuse measures 10 cm because $6^2+8^2=36+64=100=10^2$.
Alternatively, draw the 10 cm base and intersect arcs of radii 6 cm and 8 cm.
**Right triangle with legs 6 cm,8 cm and hypotenuse 10 cm.**

**3. Construct AB=5.3 cm, ∠A=30°, ∠B=120°; draw the locus equidistant from A and B.**
Draw AB=5.3 cm. Construct rays at 30° from A and 120° from B on the same side; their intersection is C.
The third angle is 30°, so BC=AB=5.3 cm and $AC=5.3\sqrt{3}\approx9.18$ cm.
Construct the perpendicular bisector of AB; this is the required locus.
**Triangle constructed; locus=perpendicular bisector of AB.**

**4. Construct a triangle with DE=7.3 cm, ∠D=42°, EF=5.4 cm.**
Draw DE=7.3 cm and a 42° ray from D. Draw circle(E,5.4 cm); it intersects the ray at two positions $F_1$ and $F_2$.
The altitude from E to the ray is $h=7.3\sin42°\approx4.885$ cm. Since $h<5.4<7.3$, both intersections are valid.

**5. Construct YX=8 cm, YZ=7 cm, XZ=6.5 cm; draw the locus equidistant from XY and XZ.**
Draw YX=8 cm. Intersect circle(Y,7 cm) and circle(X,6.5 cm) to locate Z; join YZ and XZ.
Bisect ∠YXZ. The internal bisector is the locus inside the angle; the external bisector also applies to the infinite lines.
**Required locus=angle bisector(s) at X.**

**6. Construct FG=GH=6.4 cm with ∠G=122°; intersect the loci equidistant from F,G and from FG,GH.**
Draw GF=6.4 cm and GH=6.4 cm along rays forming 122°. Join FH; base angles are 29° each.
(a) Construct the perpendicular bisector of FG
(b) Construct the internal bisector of ∠FGH.
Their intersection is the required point X; the external bisector gives a separate full-line intersection if requested.

**7. Houses Q,R are 73 m apart; scale 1 cm=10 m. Construct loci 32 m from Q and 48 m from line QR.**
Draw QR=7.3 cm. For (i), draw circle(Q,3.2 cm).
For (ii), draw two lines parallel to QR at perpendicular distance 4.8 cm on either side.
If both conditions are intended simultaneously, no intersections exist because $4.8>3.2$ cm: each parallel is farther from Q than the circle radius.
**Separate loci=circle and two parallels; no point satisfies both simultaneously .**

**8. Rectangle ABCD has AB=70 m, BC=60 m; scale 1 cm=10 m. Show the interior region less than 30 m from C and farther than 25 m from AB.**
Draw rectangle 7 cm×6 cm with AB horizontal. Draw a dashed line parallel to AB at distance 2.5 cm inside the rectangle; select the side farther from AB.
Draw a dashed quarter-circle centred at C with radius 3 cm inside the rectangle; select its interior for distance<30 m.
The required region is the intersection above the 2.5 cm line and inside the 3 cm quarter-circle. Strict inequalities exclude both boundaries.
**Shaded intersection of the interior quarter-disk and the region farther than 25 m from AB; boundaries dashed.**

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